UT Austin EDS — Developer Guide

Date: April 2026

Adobe

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UT Austin EDS — Developer Guide

Project: University of Texas at Austin
Platform: AEM Edge Delivery Services (xwalk)
Repository: `github.com/prajwal85/eds-agent-demo`
Total Blocks: 35 (8 custom UT Austin + 10 retail legacy + 17 standard)
Date: April 2026

Architecture Overview

Component	Technology
Project Type	xwalk (Universal Editor authoring)
Source CMS	Drupal 11 (migrated from utexas.edu)
Delivery	AEM Edge Delivery CDN
Author	`author-p11300-e47725.adobeaemcloud.com`
Preview	`main--eds-agent-demo--prajwal85.aem.page`
Live	`main--eds-agent-demo--prajwal85.aem.live`
Library	`main--sta-xwalk-boilerplate--aemysites.aem.page`

Repository Structure

```
/
├── blocks/                        # All block implementations (35 blocks)
│   ├── hero-video/               # Homepage video hero (custom)
│   ├── hero-banner/              # Interior page hero (custom)
│   ├── cards-article/            # News story cards (custom)
│   ├── sticky-panels/            # Sticky scroll panels (custom)
│   ├── columns/                  # Vanilla columns (enhanced with auto-detection)
│   ├── columns-promo/            # Promotional callout (custom)
│   ├── columns-impact/           # Impact stats sections (custom)
│   ├── columns-outro/            # Outro with video (custom)
│   ├── columns-resource/         # Resource link lists (custom)
│   ├── columns-featured/         # Featured columns (legacy WKND)
│   ├── hero/                     # Standard hero
│   ├── hero-adventure/           # Adventure hero (legacy WKND)
│   ├── cards/                    # Standard cards
│   ├── carousel/                 # Standard carousel
│   ├── carousel-hero/            # Hero carousel (legacy WKND)
│   ├── accordion/                # Standard accordion
│   ├── tabs/                     # Standard tabs
│   ├── quote/                    # Standard quote
│   ├── embed/                    # External embed
│   ├── video/                    # Video player
│   ├── table/                    # Data table
│   ├── modal/                    # Modal overlay
│   ├── form/                     # Form
│   ├── fragment/                 # Reusable fragment
│   ├── search/                   # Search
```

```
│   ├── header/                  # Site header
│   ├── footer/                  # Site footer
│   └── retail-*/                # Retail blocks (10, legacy)
├── styles/
│   └── styles.css               # Global styles + UT Austin brand tokens
├── scripts/
│   ├── scripts.js               # Core EDS scripts
│   └── delayed.js               # Deferred scripts
├── tools/importer/              # Import infrastructure
│   ├── import-homepage.js       # Homepage import
│   ├── import-interior.js       # Interior pages (45 pages)
│   ├── import-resource-hub.js   # Resource hub (3 pages)
│   ├── parsers/                 # Block parsers (9 files)
│   └── transformers/            # Page transformers (4 files)
├── content/                     # Migrated content + media
├── component-definition.json    # Universal Editor block definitions
├── component-models.json        # Block field models
├── component-filters.json       # Block container filters
├── .migration/project.json      # Migration config
└── Instructions.md              # Complete migration reference
```

UT Austin Custom Blocks (8)

hero-video

Files: `blocks/hero-video/hero-video.js`, `.css`, `_hero-video.json`

Used on: Homepage only

Selector: `.block-coresite-homepagehero`

Model (3 fields, 2 cells after imageAlt collapse):

Field	Type	Maps to
image	reference	Row 1 — poster/fallback image
imageAlt	text (collapsed)	—
video	reference	Row 2 — .mp4 link
text	richtext	Row 3 — headline (H1)

JS Behavior:

1. Detects `.mp4` link in Row 2 (video field) via `isVideoUrl()` helper
2. Creates `<video autoplay muted loop playsinline>` with poster from Row 1 image
3. Removes video row, leaving 2 rows: background + text overlay
4. Handles AEM media hash URLs (`/media_*`) and direct `.mp4` paths

CSS Pattern:

- `div:first-child` — absolute positioned, fills block (z-index: 0)
- `div:last-child` — relative positioned text with `linear-gradient(transparent, rgb(0 0 0 / 50%))` overlay (z-index: 1)
- H1: white, uppercase, 4rem desktop / 2.5rem mobile

hero-banner

Files: `blocks/hero-banner/hero-banner.js`, `.css`, `_hero-banner.json`
Used on: 30+ interior pages (section landing, about, academics, campus life, energy, research)
Selector: `.block-bundle-utexas-hero`

Model:

Field	Type	Maps to
image	reference	Background image
imageAlt	text (collapsed)	—
text	richtext	Heading + description

CSS: Same absolute-background pattern as hero-video but without video support. Min-height 300px desktop.

cards-article

Files: `blocks/cards-article/cards-article.js`, `.css`, `_cards-article.json`, `metadata.json`
Used on: Homepage
Selector: `.block-coresite-stories`

Model (per card row):

Field	Type	Maps to
image	reference	Card image
text	richtext	Linked headline (H3 with <a>)

CSS Grid Layout (desktop ≥900px):

```
grid-template-columns: 1fr 1fr
grid-template-rows: 1fr 1fr

li:first-child → grid-column: 2, grid-row: 1/3 (prominent, right)
li:nth-child(2) → grid-column: 1, grid-row: 1 (secondary, left top)
li:nth-child(3) → grid-column: 1, grid-row: 2 (secondary, left bottom)
```

- Prominent card: full-height image + absolute-positioned headline with gradient
- Secondary cards: CSS grid `grid-template-columns: 1fr 1fr` (image left, text right)
- Background: `var(--burnt-orange, #bf5700)`

sticky-panels

Files: `blocks/sticky-panels/sticky-panels.js`, `.css`, `_sticky-panels.json`, `metadata.json`
Used on: Homepage (3 consecutive instances)
Also auto-detected in: `columns` block when pattern matches

Model (2 cells per instance):

Field	Type	Maps to
image	reference	Portrait image (left sticky column)
imageAlt	text (collapsed)	—
content	richtext	Heading + paragraph + CTA + stats

JS Behavior:

1. Adds `.sticky-panels-panel`, `.sticky-panels-image`, `.sticky-panels-content` classes
2. Detects `<p><strong>` elements → marks as `.sticky-panels-stat`
3. Detects first `<a>` → marks as `.sticky-panels-cta`
4. IntersectionObserver (threshold 0.3) adds `.sticky-panels-visible` for fade-in animation

CSS (desktop ≥900px):

```
.sticky-panels-image {
  position: sticky;
  top: var(--nav-height, 64px);
  height: calc(100vh - var(--nav-height, 64px));
  flex: 0 0 40%;
}
.sticky-panels-content { flex: 1; }
.sticky-panels-stat { opacity: 0; transform: translateY(20px); transition: 0.6s; }
.sticky-panels-stat.sticky-panels-visible { opacity: 1; transform: none; }
```

columns (enhanced vanilla)

Files: `blocks/columns/columns.js`, `.css`
Used on: AEM Author delivers impact/outro sections as `columns`

Enhanced JS with auto-detection:

```
// 1. Sticky Panels Detection
if (col1 has image AND col2 has <strong> stats + <a> CTA)
  → applyStickypanels(block)

// 2. Video Detection
if (any column has link containing .mp4 or /media_)
  → creates autoplay muted loop video

// 3. Standard columns behavior (fallback)
  → adds columns-img-col class to picture-only columns
```

Why needed: AEM Author page was authored with vanilla `columns` blocks before custom block names existed. The JS auto-detects content patterns and applies the correct decoration.

columns-promo

Files: `blocks/columns-promo/columns-promo.js`, `.css`, `_columns-promo.json`, `metadata.json`
Used on: Homepage (UT Dell Campus promo)
Selector: `#block-coretheme-homepagemidpagepromo`

Standard 2-column layout. Used with section-metadata `style: burnt-orange`.

columns-impact

Files: `blocks/columns-impact/columns-impact.js`, `.css`, `_columns-impact.json`, `metadata.json`
Used on: Homepage (3 instances — imported content)
Selector: `.block-coresite-impact-1/2/3`

Special handling in parser:

- Rewrites image URLs from `1170x1536` to `585x768` (mapped versions)
- `KNOWN_STATS` fallback: stats extracted by heading text when JSDOM displaces `.stats` elements

columns-outro

Files: `blocks/columns-outro/columns-outro.js`, `.css`, `_columns-outro.json`, `metadata.json`
Used on: Homepage ("For Texas, For the Future")
Selector: `.block-homepage-outro`

JS: Detects `.mp4` links → creates autoplay video. Adds `.columns-outro-video-col` class.

columns-resource

Files: `blocks/columns-resource/columns-resource.js`, `.css`, `_columns-resource.json`, `metadata.json`
Used on: Resource hub pages (3) + most interior pages (35+)
Selector: `.utexas-layout--twocol-wrapper`, `.utexas-layout--threecol-wrapper`

Most common block. Extracts heading + link lists from Drupal two-column wrappers.

Standard Blocks (17)

Available from the EDS Block Collection, already in the project:

Block	Directory	Description
hero	`blocks/hero/`	Standard hero (text + image)
cards	`blocks/cards/`	Standard card grid
columns	`blocks/columns/`	Side-by-side columns (enhanced)
accordion	`blocks/accordion/`	Expandable Q&A sections
tabs	`blocks/tabs/`	Tabbed content panels
carousel	`blocks/carousel/`	Rotating slideshow
quote	`blocks/quote/`	Testimonial/pullquote
embed	`blocks/embed/`	External embed (YouTube, etc.)
video	`blocks/video/`	Video player (YouTube, Vimeo, MP4)
table	`blocks/table/`	Data table
modal	`blocks/modal/`	Modal overlay
form	`blocks/form/`	Form inputs
fragment	`blocks/fragment/`	Reusable content
search	`blocks/search/`	Search functionality
header	`blocks/header/`	Site navigation header
footer	`blocks/footer/`	Site footer
carousel-hero	`blocks/carousel-hero/`	Hero carousel variant (WKND)

Legacy Blocks (10 — Retail)

From previous WKND project, not actively used for UT Austin:

`retail-hero`, `retail-cards`, `retail-featured`, `retail-home-featured`, `retail-home-flash-sales`, `retail-newsletter`, `retail-product`, `retail-store`, `retail-testimonials`, `retail-tips`

Design Tokens (styles/styles.css)

```
:root {
  /* UT Austin brand colors */
  --link-color: #bf5700;              /* Burnt orange */
  --link-hover-color: #9d4700;        /* Darker orange */
  --dark-color: #333f48;              /* Charcoal */
  --text-color: #212529;              /* Body text */
  --heading-color: #333f48;           /* Headings */
  --burnt-orange: #bf5700;            /* Brand alias */
  --charcoal: #333f48;               /* Brand alias */
  --background-color: white;
  --light-color: #f8f8f8;
  --overlay-background-color: #ebe7e1;

  /* Fonts */
  --body-font-family: roboto, roboto-fallback, sans-serif;
  --heading-font-family: roboto-condensed, roboto-condensed-fallback, sans-serif;

  /* Buttons: 4px radius, orange border, solid fill */
  /* Images: width 100%, height auto (no circular cropping) */
}
```

Import Infrastructure

Import Scripts

Script	Template	Pages	Blocks Used
`import-homepage.js`	Homepage	1	hero-video, cards-article, columns-promo, columns-impact, columns-outro
`import-resource-hub.js`	Resource Hub	3	columns-resource
`import-interior.js`	All other	45	hero-banner, columns-resource

Parsers (`tools/importer/parsers/`)

Parser	Block	Source Selector
`hero-video.js`	hero-video	`.block-coresite-homepagehero`
`hero-banner.js`	hero-banner	`.block-bundle-utexas-hero`
`cards-article.js`	cards-article	`.block-coresite-stories`
`columns-impact.js`	columns-impact	`.block-coresite-impact-1/2/3`
`columns-promo.js`	columns-promo	`#block-coretheme-homepagemidpagepromo`
`columns-outro.js`	columns-outro	`.block-homepage-outro`
`columns-resource.js`	columns-resource	`.utexas-layout--twocol-wrapper`

Transformers (`tools/importer/transformers/`)

Transformer	Hook	Purpose

`utexas-cleanup.js`	before + after	Remove header/footer/nav, convert blockquotes, insert section-metadata
`utexas-sections.js`	before	Insert section breaks while DOM elements exist
`wknd-cleanup.js`	—	Legacy WKND (unused)
`wknd-sections.js`	—	Legacy WKND (unused)

Re-importing Pages

```
SCRIPTS="/home/node/.excat-marketplace/excat/skills/excat-content-import/scripts"

# Re-bundle after changes
"$SCRIPTS/aem-import-bundle.sh" --importjs tools/importer/import-interior.js

# Re-import single page
echo "https://www.utexas.edu/about-texas" > /tmp/url.txt
node "$SCRIPTS/run-bulk-import.js" \
  --import-script tools/importer/import-interior.bundle.js \
  --urls /tmp/url.txt

# Re-import all interior pages
node "$SCRIPTS/run-bulk-import.js" \
  --import-script tools/importer/import-interior.bundle.js \
  --urls tools/importer/urls-interior.txt
```

Known Issues & Workarounds

1. Stats Not Extracted from Impact Sections

Cause: JSDOM displaces `.stats` elements outside parent due to `<picture>` with empty `<source>` tags.
Fix: `columns-impact.js` parser uses `KNOWN_STATS` fallback matched by heading text.

2. Video on AEM Cloud

Cause: Relative paths don't resolve on AEM delivery; URL format differs.
Fix: `isVideoUrl()` helper checks `.mp4` anywhere in path AND AEM `/media_` hash URLs. Absolute `/content/media/` paths used.

3. Sticky Panels on AEM Author

Cause: AEM Author delivers as `columns` (not `sticky-panels`).
Fix: `columns.js` auto-detects pattern (image + stats + CTA) and applies sticky-panels decoration.

4. xwalk/max-cells Lint Rule

Cause: Maximum 4 cells per block model.
Fix: Use 2 cells per sticky-panels instance. 3 consecutive blocks for 3 panels.

5. Circular Images

Cause: Global CSS `main p > img:only-child { border-radius: 50% }` from boilerplate.
Fix: Removed circular styling, replaced with `width: 100%; height: auto`.

6. Blockquotes Passing Through

Cause: Raw `<blockquote>` from Drupal not handled.

Fix: `utexas-cleanup.js` transformer converts to EDS quote blocks in `afterTransform`.

Development Commands

```
# Start local dev server
aem up

# Run all linting
npm run lint

# CSS lint only
npm run lint:css

# JS lint only
npm run lint:js

# Preview homepage
open http://localhost:3000/content/index
```

Deployment Flow

```
Developer pushes to main
    → GitHub Actions: npm run lint (JS + CSS + xwalk rules)
    → AEM Code Sync detects changes
    → Preview updates at .aem.page
    → Author publishes via Sidekick
    → Live at .aem.live
```

Lint Rules Summary

Rule	File Type	Requirement
`xwalk/max-cells`	JSON	Max 4 cells per block model
`no-unused-vars`	JS	No unused variables
`color-function-notation`	CSS	Modern `rgb()` notation
`alpha-value-notation`	CSS	Percentage (e.g., `60%` not `0.6`)
`color-function-alias-notation`	CSS	`rgb` not `rgba`

Key Files Reference

File	Purpose

`fstab.yaml`	AEM Author mountpoint
`.migration/project.json`	Migration config (type: xwalk, libraryUrl)
`component-definition.json`	Block definitions for Universal Editor
`component-models.json`	Block field models (max 4 cells each)
`component-filters.json`	Block container filters
`Instructions.md`	Complete migration reference (DOM patterns, debugging tips)
`tools/importer/page-templates.json`	10 template definitions with selectors